# INOVEX.

Operations Sentinel v2.0 — Complete Standard Operating Manual

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OPERATIONAL SPEC: 2.0.4
APPLICABLE: FOUNDERS & OPERATORS
CLASSIFICATION: INTERNAL STRICT
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# 1. System Overview & Architecture

**INOVEX Operations Sentinel v2.0** is a high-assurance operations cockpit and zero-latency telemetry enclave designed for high-performance engineering teams. It bridges autonomous AI task dispatching with real-time biometric telemetry verification, 90-minute focused burst checkpoints, and screen tracking proof of work.

> **Core Operating Philosophy: Zero Noise, Zero Delay**
> INOVEX replaces status meetings and micromanagement with continuous autonomous telemetry. Founders maintain high-level visibility across engineering nodes without interrupting flow state.

## System Architecture & Access Tiers

| Access Tier | Permissions & Scope | Primary Interfaces |
|---|---|---|
| `EXECUTIVE FOUNDER` | Global read/write access. Autonomous AI task dispatching, pacing telemetry graph inspection, deep 90-minute video anomaly audits, and node authorization management. | Orchestration Grid, AI Dispatch Terminal, Health Telemetry (Waveform Graph & Tabular Matrix), Video Audits Enclave. |
| `ENGINEERING OPERATOR` | Scoped workspace execution. Micro-sprint timer tracking, workstream directive adherence, sub-second telemetry broadcast, and proof-of-work deliverable submission. | Sprint Workspace, Fixed 90-min Countdown Timer, Screen Sentinel Monitor, Deliverable Submission Webhook. |

# 2. First-Time Setup & Node Registration

## Initial Founder Account Creation (Bootstrap Phase)

### Step 2.1 — Accessing Initial Setup

When the system is freshly deployed and contains zero registered accounts, visiting the production portal directly displays the **System Initial Setup** gateway.

- Navigate to the application production URL: `https://dist-omega-rosy-35.vercel.app`
- Click on the **Register New Node** tab.
- The role defaults to **Executive Founder**.
- Enter your Full Name, official Email Address, Specialization (e.g., *CEO / Technical Director*), and a secure password (minimum 4 characters).
- Click **Register Node & Sign In**.

*Security Note: Once the first Founder account is initialized, unauthenticated public Founder self-registration is permanently disabled.*

## Employee Node Registration

### Step 2.2 — Onboarding Engineering Team Members

- New employees navigate to the same production URL and select **Register New Node**.
- Select the **Engineering Operator** tab.
- Input Name, Email, and Technical Skill Set (e.g., *Pipeline Eng, Agentic Eng, Backend Systems*).
- The system automatically computes a unique **Hardware MAC Signature** for device binding.
- Set a password and submit. The employee node is immediately registered and persisted in the cloud enclave.

# 3. Founder Operational Guide

The Founder Dashboard provides end-to-end command and oversight across all dispatched micro-sprints and connected engineering workstations.

## 3.1 The Orchestration Grid

The top section displays active telemetry specimens across your team:

- **Total Nodes**: Breakdown of registered Executive Founders and active Engineering Operators.
- **Active Sprints**: Real-time counter of micro-sprints currently in execution.
- **Average Health**: Aggregate pacing health index (0–100%) calculated dynamically across active workstations.
- **Zero-Latency Status**: Real-time indicator displaying the live Sub-Second connection via Supabase Realtime.

## 3.2 Autonomous Sprint Dispatcher (Action 01)

The Autonomous Multi-Agent Dispatcher converts high-level objectives into executable 90-minute burst deliverables:

### How to Dispatch Work

1. In the **Autonomous Sprint Dispatcher** box, type the requirements, technical constraints, or user story into the prompt area.
2. Click **Initialize AI Dispatch**.
3. The 4-stage pipeline executes automatically:
    - **01 Specs Ingest**: Normalizes requirements into discrete deliverables.
    - **02 Deconstruct**: Evaluates complexity and estimated velocity.
    - **03 Capacity Match**: Intelligently routes the assignment to the least-loaded qualified node.
    - **04 Checkpoint Webhook**: Issues a 90-minute deadline timer directly to the employee's workstation.

## 3.3 Health Telemetry: Waveform Graph & Tabular Matrix (Action 02)

Founders can toggle between two continuous telemetry perspectives:

- **Graph View (Default)**: Renders a real-time responsive SVG Waveform chart showing team pacing velocity, keystroke cadence, and sub-second hardware integrity. Nodes with stable input entropy display smooth green waveforms, while idle or divertive sessions display downward dips.
- **Tabular Format**: Provides a tabular breakdown of each dispatched task, assigned node, active window focus (e.g. *VS Code, Terminal*), and numerical AI pacing score.

## 3.4 Video Audits & AI Anomaly Inspection (Action 03)

When an employee completes a 90-minute burst, their screen recording is processed through the AI Inspector. Founders can:

- Click **Inspect Audit** on any completed session card.

- Review the **Neural Focus Breakdown** (e.g. *88% Active Development, 8% Documentation, 4% Flagged/Idle*).
- Click on specific **Flagged Time-Offset Bookmarks** (e.g. *36:00 Off-Task Application Shift*) to jump directly to the video timestamp.
- Review the AI Sentinel Verdict ( `CLEAN_VERIFIED` or `CAUTIONARY_FLAGGED` ).

# 4. Employee Workspace Guide

The Employee Workspace is designed for deep, uninterrupted engineering focus. It centers around strict 90-minute burst sessions.

## 4.1 The 90-Minute Sprint Cycle

### Sprint Protocol

1. **Task Inspection**: Review assigned sprint title, architectural brief, and delivery deadline in the left-hand rail.
2. **Initiate Sprint Burst**: Click **Initiate 90-Min Sprint**. The fixed countdown timer begins at `90:00` and counts down to zero.
3. **Screen Sentinel Activation**: Screen tracking activates to record the session. A live stream preview appears in the workspace monitor card.
4. **Focus Directives**: Keep technical tools (IDE, Terminal, API Docs) in active focus. Avoid non-work diversions to keep pacing scores nominal (85%+).

## 4.2 Submitting Proof of Work (PoW)

When engineering objectives are achieved:

- In the **Proof of Work & Deliverable Submission** section, enter the deliverable reference (e.g., Pull Request URL, Git commit hash, or staging link).
- Add technical implementation notes detailing modifications and test coverage.
- Click **Submit Deliverable & Request Verification**.
- The sprint transitions to `review` status and the screen recording is automatically packaged for Founder verification.

## 4.3 AI Sentinel Behavioral Warnings

If an off-task application is sustained for more than 45 seconds, the automated sentinel issues an in-app cautionary prompt:

- **Warning 1/2**: Cautionary alert notifying the operator of off-task divergence.
- **Warning 2/2**: Secondary flag. If unaddressed, the sprint session terminates automatically to protect audit provenance.

# 5. Security, Hardware Anti-Spoofing & Offline Resilience

## 5.1 Hardware Anti-Spoofing Sentinel

INOVEX monitors input entropy (keystroke timing jitter and mouse coordinate distributions). If automated macro replay bots or hardware mouse jigglers are introduced, the sentinel flags a `HARDWARE_SPOOF_ALERT`, alerting the

Founder enclave immediately.

## 5.2 Dual-Mode Cloud Failover

The web client utilizes dual-mode resilience. In cloud-only deployment (Vercel + Supabase), state sync, authentication, and task transitions route with sub-15ms latency directly through Supabase Postgres and Realtime Channels. No local server maintenance or database hosting costs are required.

## 5.3 PWA & Android Installation

> **PWA Android Installation**
>
> Open `https://dist-omega-rosy-35.vercel.app` in Google Chrome on your Android smartphone. Tap **Install App** or select **Add to Home Screen** from the browser menu to install INOVEX as a native fullscreen Android application.